Formulation Development, Physicochemical Characterization, and In Vitro Drug Release Kinetics of Curcumin-Loaded PLGA Nanoparticles for Enhanced Oral Drug Delivery

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Abstract

Background: Curcumin, a naturally occurring polyphenolic compound derived from *Curcuma longa*, exhibits promising antioxidant, anti-inflammatory, and therapeutic properties. However, its pharmaceutical applications are limited by poor aqueous solubility, chemical instability, rapid metabolism, and low oral bioavailability. Poly(lactic-co-glycolic acid) (PLGA), a biodegradable and biocompatible polymer, represents a promising nanocarrier system for improving curcumin encapsulation, dissolution characteristics, and controlled drug delivery.

Objective: The present conceptual study aimed to design, optimize, and characterize curcumin-loaded PLGA nanoparticles and systematically evaluate the influence of polymer concentration, PLGA composition, stabilizer concentration, and freeze-drying on nanoparticle characteristics and *in vitro* drug release kinetics.

Methods: Six proposed nanoparticle formulations (F1–F6) were developed using nanoprecipitation and high-speed homogenization approaches, with variations in PLGA concentration, lactide-to-glycolide ratio, and polyvinyl alcohol (PVA) stabilizer concentration. Freeze-drying was incorporated to investigate its potential effects on nanoparticle stability and drug-release behavior. Formulation characterization included particle size analysis by dynamic light scattering (DLS), polydispersity index, zeta potential, drug loading, encapsulation efficiency, and high-performance liquid chromatography (HPLC)-based drug quantification. A comparative 24-hour *in vitro* dissolution study was modeled using a numerical proof-of-concept dataset. Drug-release profiles were evaluated using zero-order, first-order, Higuchi, Korsmeyer–Peppas, and Weibull kinetic models to investigate potential release mechanisms and formulation-dependent differences.

Results: The conceptual formulation model targeted nanoparticle sizes of approximately 160–190 nm, encapsulation efficiencies exceeding 85%, and sustained curcumin release over 24 hours. Comparative release profiles illustrated how changes in polymer concentration, PLGA composition, and stabilizer concentration could influence the initial burst release, cumulative drug release,

and subsequent release kinetics. Higher polymer concentrations were hypothesized to reduce drug diffusion rates through increased matrix density, whereas changes in PLGA composition were expected to influence hydration, diffusion, and polymer degradation behavior. Kinetic modeling provided a framework for comparing diffusion-controlled, concentration-dependent, and anomalous transport mechanisms. The generated release profiles and kinetic parameters represent constructed proof-of-concept examples rather than experimentally measured results.

Conclusion: The proposed study demonstrates a systematic formulation-development strategy for investigating curcumin-loaded PLGA nanoparticles as potential controlled-release oral drug delivery systems. The comparative formulation approach and mathematical kinetic analysis provide a framework for understanding polymer-dependent drug-release behavior and identifying promising formulations for further optimization. Experimental validation through physicochemical characterization, HPLC-based release testing, stability studies, and oral absorption assessments will be necessary to establish formulation performance and potential bioavailability improvements.

Keywords: Curcumin; PLGA Nanoparticles; Nanoprecipitation; Oral Drug Delivery; Encapsulation Efficiency; Controlled Drug Release; HPLC; Higuchi Model; Korsmeyer–Peppas Model; Drug Release Kinetics; Formulation Optimization.